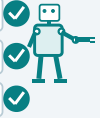
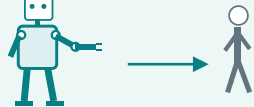


ASIMOV CASCADE

A stale contact estimate crosses planning and balance control while every module remains locally correct.

STAGE	A COMPONENT CHECKS C1 · LOCAL CONTRACTS	B SYSTEM INTEGRATION COUPLED ASSUMPTIONS	C CONTEXT SHIFT C2 · INTERFACE MISMATCH	D CASCADING FAILURE C3 PROPAGATES · C4 UNSAFE
LOGIC	Each subsystem satisfies its own specification under the assumed operating context.	Interfaces coordinate through shared timing, state and mode assumptions.	The world changes while an interface continues to supply an outdated state.	The mismatch crosses at least two interfaces and alters the system trajectory.
WORKED EXAMPLE	Perception Planning Control  All local tests pass.	 LIVE STATE Current state is shared.	 STALE ESTIMATE Person moves; shared state is stale.	Stale contact Whole-body plan Balance control UNSAFE TRAJECTORY The stale state changes the whole path.

DIAGNOSTIC BOUNDARY: fresh-state replanning contains the mismatch (interface fault); downstream reuse propagates it (cascade). Harm also requires exposure.